

SIMATS ENGINEERING (SAVEETHA SCHOOL OF ENGINEERING)

Department of Artificial Intelligence and Machine Learning

ASSESSMENT TOOL 2 – Scenario-Based Assignment (5 QUESTIONS)

Course Code:	MLA0407	Course Title:	Deep Learning
CO Assessed:	CO2	Assessment:	Assessment Tool 2 – Scenario-Based Assignment
Weightage:	35% of CIA	Total Marks:	(5 Questions × 10 Marks)
CO1 Statement:	<i>learning algorithms using perceptrons, forward propagation, backpropagation, gradient descent optimization techniques, and Maximum Likelihood Estimation for solving classification problems.</i>		<i>fundamental concepts of machine learning and neural networks to design and analyze</i>
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Section / Batch:	2024-AI & ML	Date:	7.08.2026

Code of Conduct

I S. Likiteswar Reddy(192425120)(NAME/ REG NO) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment. I understand that any violation of academic integrity rules will result in disciplinary action. Signature of the Student: S.Likiteswar Reddy

QUESTIONS: 5 Total Marks: 50

Annexure A

Scenario-Based Assignment Duration: 1hr

Deep Learning – Assessment 2

1. Machine Learning vs Deep Learning

A healthcare company wants to develop an AI system to detect pneumonia from chest X-ray images. Initially, the development team plans to use traditional Machine Learning with manually extracted image features. However, after collecting millions of medical images, they decide to switch to a Deep Learning approach.

Question:

Compare Machine Learning and Deep Learning in the context of this scenario. Justify why Deep Learning is a better choice for this application by discussing data requirements, feature extraction, computational resources, accuracy, and real-world implementation challenges.

Answer:

- **Machine Learning (ML):** Relies on manually engineered features, works with small datasets, limited accuracy in complex medical imaging.
- **Deep Learning (DL):** Learns features automatically from raw pixels, requires large datasets and GPUs, achieves higher accuracy.
- **Justification:** With millions of chest X-rays, DL is superior because CNNs can capture subtle pneumonia patterns, eliminating manual feature extraction. Despite higher computational cost, DL provides better scalability, accuracy, and robustness — essential in healthcare diagnostics.

2. Representation Learning

A retail company has millions of customer purchase records and wants to automatically identify customer buying patterns without manually selecting important features. The company plans to use a Deep Learning model that can learn meaningful representations directly from raw transactional data.

Question:

Explain the concept of Representation Learning and discuss how it helps solve the company's problem.

Compare it with traditional feature engineering and explain its advantages in handling complex real world datasets.

Answer:

- **Concept:** Representation Learning enables models to learn meaningful features directly from raw data.
- **Traditional Feature Engineering:** Manual, time-consuming, and biased.
- **Advantages:** Handles complex, high-dimensional datasets, automatically discovers hidden buying patterns, reduces human bias.
- **Application:** The retail company can identify customer segments and preferences without manually selecting features, improving recommendation and marketing strategies.

3. Width and Depth of Neural Networks

A startup is developing a handwritten digit recognition system. Two engineers propose different neural network architectures. One suggests using a very wide network with a single hidden layer containing thousands of neurons, while the other recommends a deeper network with multiple hidden layers and fewer neurons per layer.

Question:

Analyze both approaches by explaining the concepts of width and depth in neural networks. Discuss how each architecture affects feature learning, computational complexity, model performance, and generalization. Recommend the most suitable architecture with proper justification.

Answer:

- **Wide Network:** One hidden layer with thousands of neurons, risk of overfitting, poor generalization, high computational cost.
- **Deep Network:** Multiple hidden layers with fewer neurons, learns hierarchical features (edges → shapes → digits), better generalization.
- **Recommendation:** A deeper network is more suitable for digit recognition because it captures hierarchical structures efficiently and balances accuracy with computational complexity.

4. Activation Functions

A company is building a Deep Learning model for image classification. During training, the team observes problems such as slow convergence, vanishing gradients, dead neurons, and unstable outputs in the final classification layer.

Question:

Evaluate the suitability of Sigmoid, Tanh, ReLU, Leaky ReLU, ELU, and Softmax activation functions for different layers of the neural network. Explain which activation function should be used in each stage of the model and justify your choices based on their characteristics.

Answer:

- **Sigmoid/Tanh:** Cause vanishing gradients, slow convergence.
- **ReLU:** Fast convergence, widely used, risk of dead neurons.
- **Leaky ReLU/ELU:** Solve dead neuron issue, stabilize training.
- **Softmax:** Best for final classification layer (outputs probabilities).
- **Recommendation:** Use ReLU/Leaky ReLU/ELU in hidden layers for efficient training, and Softmax in the output layer for classification.

5. Unsupervised Training of Neural Networks

A social media platform collects billions of unlabeled user images every month. Since manually labeling such a massive dataset is expensive and time-consuming, the company wants to train a neural network that can automatically discover hidden patterns and useful representations. **Question:**

Explain how unsupervised training of neural networks can address this problem. Discuss the learning process, advantages over supervised learning in this scenario, and describe suitable techniques that can be applied for representation learning.

Answer:

- **Process:** Learns hidden structures without labeled data.
- **Advantages:** Saves labeling cost, discovers general representations, scales to billions of images.
- **Techniques:** Autoencoders, self-supervised learning (contrastive methods), clustering with deep embeddings.
- **Application:** Enables grouping of similar images, learning embeddings for personalization, and improving recommendation systems.

6. Restricted Boltzmann Machines (RBM)

An online music streaming platform wants to improve its recommendation system by learning hidden relationships between users and songs from historical listening data without using labeled information.

Question:

Explain how a Restricted Boltzmann Machine (RBM) can be used in this recommendation system. Describe the architecture of an RBM, its working principle, training process, and discuss its advantages and limitations in real-world recommendation applications.

Answer:

- **Architecture:** Two layers — visible (users/songs) and hidden (latent features).
- **Working Principle:** Learns probability distribution of user-song interactions.
- **Training:** Contrastive Divergence algorithm.
- **Advantages:** Captures hidden relationships, useful for collaborative filtering.
- **Limitations:** Training instability, less scalable than modern DL methods.
- **Application:** Improves recommendations by modeling hidden user preferences and song affinities.

7. Autoencoders

A manufacturing company installs hundreds of sensors on industrial machines to continuously monitor their health. Most of the collected sensor data represents normal machine behavior, while faulty conditions occur very rarely. The company wants an AI system that can automatically identify abnormal machine behavior without requiring labeled fault data.

Question:

Discuss how an Autoencoder can be used for anomaly detection in this scenario. Explain the architecture of an Autoencoder, the roles of the encoder, bottleneck layer, and decoder, and describe how reconstruction error helps detect abnormal conditions.

Answer:

- **Architecture:** Encoder → Bottleneck → Decoder.
- **Roles:** Encoder compresses input, bottleneck captures key features, decoder reconstructs input.
- **Detection:** Normal data reconstructs well (low error), faulty data poorly (high error).
- **Application:** Reconstruction error signals anomalies, enabling detection of rare machine

faults without labeled fault data.

8. Deep Learning Applications

A smart city project aims to integrate multiple AI-based services, including face recognition for security, speech recognition for public information kiosks, medical image analysis in hospitals, autonomous traffic monitoring, and intelligent language translation for tourists. **Question:**

Explain how Deep Learning can be applied in each of these services. Discuss the advantages of Deep Learning in solving these real-world problems and identify the challenges that may arise during deployment.

Answer:

- **Face Recognition:** CNNs for identity verification.
- **Speech Recognition:** RNNs/Transformers for audio-to-text.
- **Medical Imaging:** CNNs for disease detection.
- **Traffic Monitoring:** Object detection for vehicles/pedestrians.
- **Language Translation:** Transformers (seq-to-seq).
- **Advantages:** High accuracy, automation, scalability.
- **Challenges:** Privacy concerns, bias, computational cost, deployment complexity.

9. Selecting the Appropriate Deep Learning Technique

A technology company is planning to build an AI-powered platform with the following features:

- Image classification for uploaded photos
- Customer recommendation system
- Data compression for cloud storage
- Feature extraction from unlabeled data

The management asks the AI team to recommend suitable Deep Learning techniques for each task.

Question:

Identify the most appropriate Deep Learning models or techniques for each feature. Justify your recommendations by explaining the working principles and advantages of Representation Learning, Autoencoders, Restricted Boltzmann Machines, and other relevant Deep Learning concepts.

Answer:

- **Image Classification:** CNNs — learn spatial features from images.
- **Recommendation System:** RBMs or Neural Collaborative Filtering — capture hidden user-item relationships.
- **Data Compression:** Autoencoders — reduce dimensionality while preserving information.
- **Feature Extraction (Unlabeled Data):** Representation Learning — self-supervised models discover useful features.
- **Justification:** Each technique aligns with the task's nature, ensuring efficiency and accuracy.

10. Designing a Deep Learning Solution

A national disaster management agency wants to build an intelligent emergency response system capable of analyzing satellite images, identifying damaged buildings, classifying disaster severity, and providing rescue recommendations. The system must process large amounts of image data accurately and efficiently.

Question:

Design a Deep Learning-based solution for this application. Explain whether Machine Learning or Deep Learning is more suitable, discuss the importance of Representation Learning, recommend appropriate activation functions for the network, and suggest whether unsupervised techniques such as Autoencoders or Restricted Boltzmann Machines could improve the overall performance of the system. Support your answer with appropriate reasoning and real-world examples.

Answer:

- **Choice:** Deep Learning is more suitable due to large satellite image datasets ,complex tasks.
- **Representation Learning:** Extracts meaningful features from raw images, improving classification and damage detection.
- **Activation Functions:** ReLU/Leaky ReLU for hidden layers, Softmax for classification.
- **Unsupervised Techniques:** Autoencoders detect unusual damage patterns; RBMs can model hidden severity relationships.
- **Real-World Example:** Similar systems are used in earthquake damage assessment and flood mapping.
- **Outcome:** Accurate, scalable, and efficient emergency response system that supports rescue operations.

Rubrics for Calculation

Criteria	Weightage	Level 4 (Exemplary)	Level 3 (Proficient)	Level 2 (Developing)	Level 1 (Beginning)
Technical Content Accuracy	30	Highly accurate, indepth, and insightful technical explanation	Technically correct content with adequate depth and clarity	Basic concepts presented with limited accuracy and depth	Content contains major technical errors; concepts poorly understood
Problem Identification	25	Problem rigorously analyzed using appropriate and advanced methods	Problem clearly defined with a logical engineering approach	Problem identified but analysis or methodology is weak	Problem statement unclear or incorrect; no logical approach
Use of Tools	20	Advanced tools, well analyzed data, and highly effective visuals	Appropriate tools and data used effectively with clear visuals	Limited use of tools and basic data interpretation	Minimal or incorrect use of tools and data; poor visuals
Communication	15	Highly engaging, confident, and audience-focused delivery	Clear, organized, and professional communication	Understandable but lacks clarity, structure, or confidence	Poor organization, unclear speech, ineffective slides
Professionalism	10	Exemplary professionalism, leadership, and ethical awareness	Professional conduct with effective team collaboration	Basic professionalism; uneven team participation	Lacks professionalism; minimal contribution or ethical awareness

Faculty In-charge Course Coordinator Program Director

Signature: _____ Signature: _____ Signature: _____ Date: _____

_____ Date: _____ Date: _____